

Practice Test III (Version 2)

Name
Date
Course

I) Write your name, the date, and the course title (i.e. Trigonometry).

II) Show all work.

III) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.

IV) You will need a calculator for this test (be sure that it is in degree mode).

V) Each problem is worth four points for a total of 100 points.

Find the values of the following expressions without a calculator.

① $\cos^{-1}(0)$

② $\csc^{-1}(-2)$

Find the value of the following expression. You will need a calculator. Round to the nearest tenth.

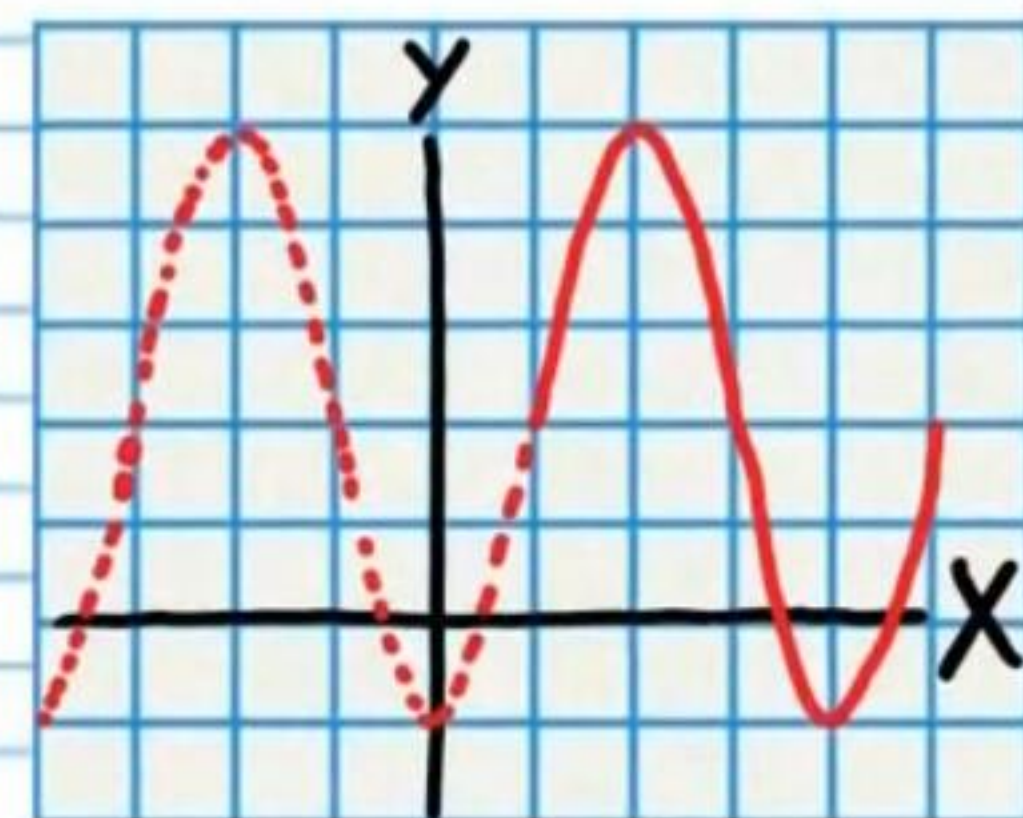
③ $\tan^{-1}(-3.67)$

Write an equation corresponding to each graph.

★ There is more than one equation that can be written for each graph (e.g. cosine functions or sine functions can represent the same graph). To avoid confusion, use the sine or cosecant functions rather than the cosine or secant functions. ★ Each horizontal unit on the graphs is 90° . Each vertical unit is one.

④ General Form of the Sine Function: $y = A \sin(\omega x - \phi) + B$

Midline Equation:



Amplitude:

Omega:

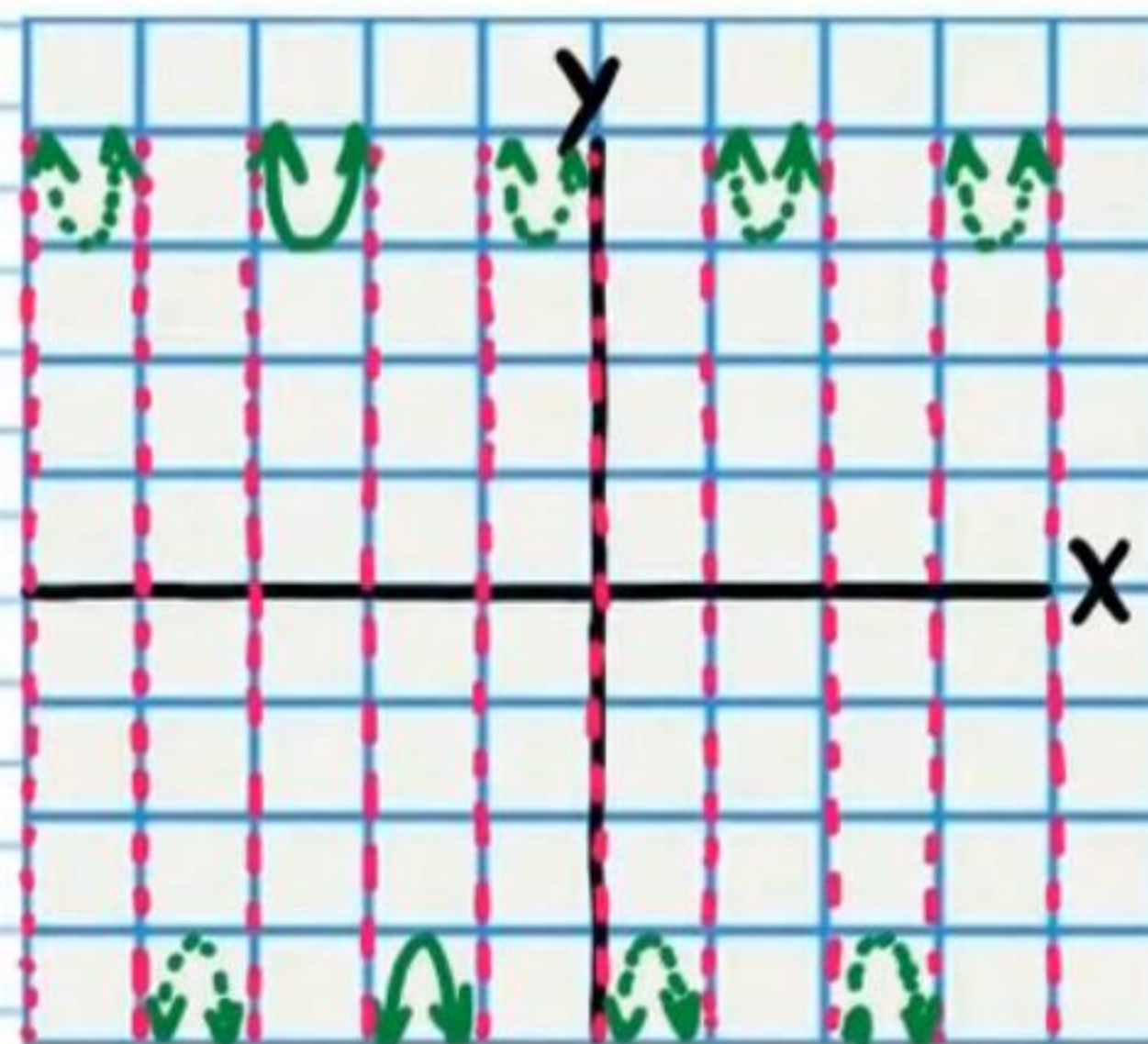
Phase Shift:

⑤ General Form of the Cosecant Function: $y = A \csc(\omega x - \phi) + B$

Midline Equation:

A:

Omega:



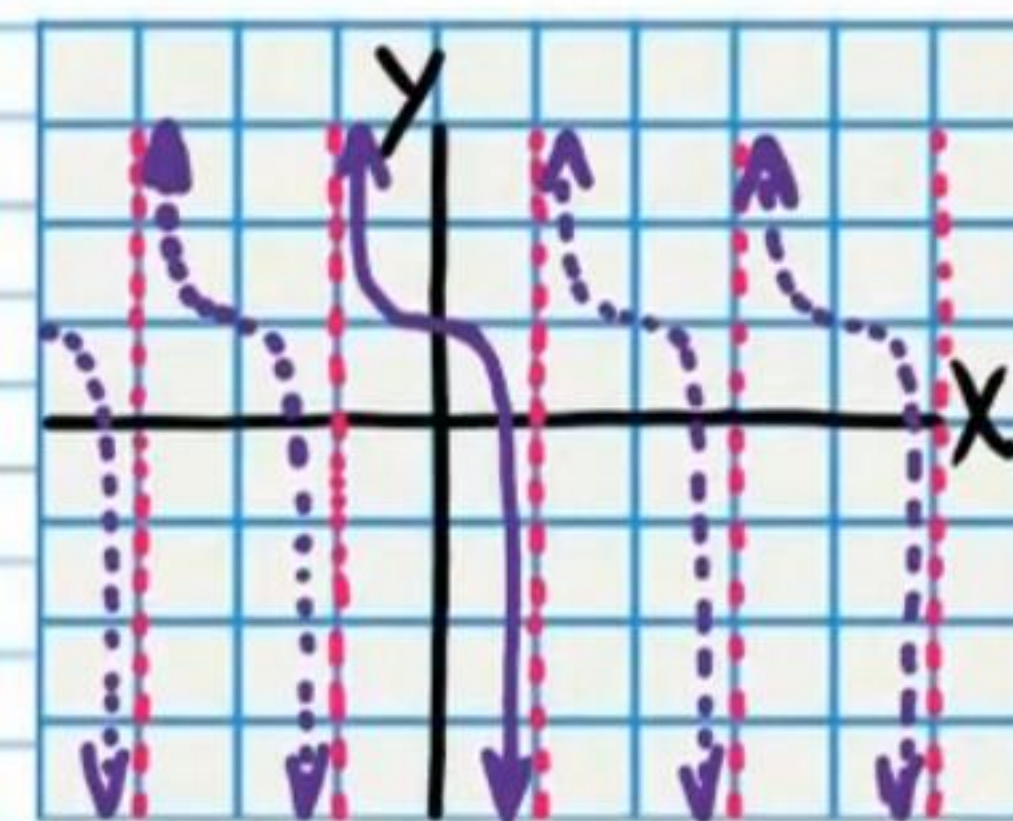
Phase Shift:

⑥ General Form of the Cotangent Function: $y = A \cot(\omega x - \phi) + B$

Midline Equation:

A: ★ Assume $A=1$.

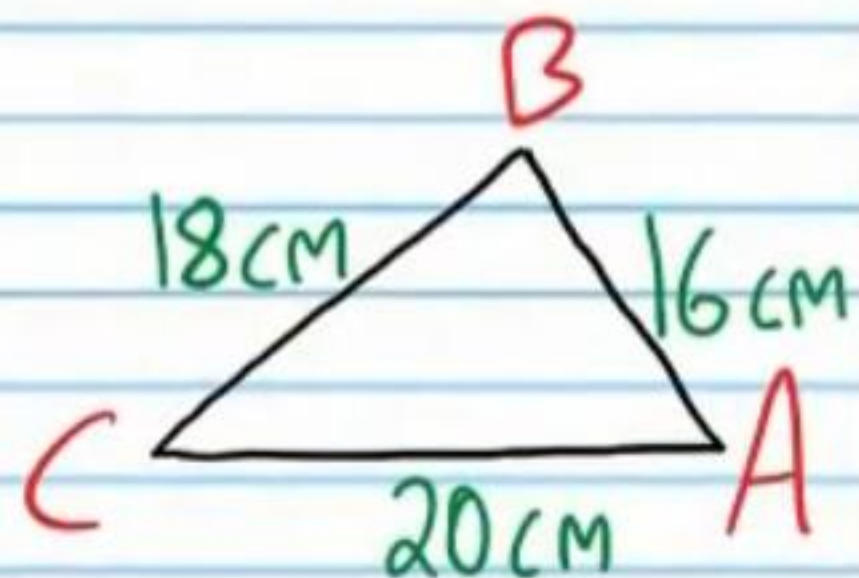
Omega: P



Phase Shift:

Find the missing side lengths and angle measures of each triangle. In other words, solve each triangle. There might not be a triangle that exists for some of these problems.

⑦ $m\angle B$.

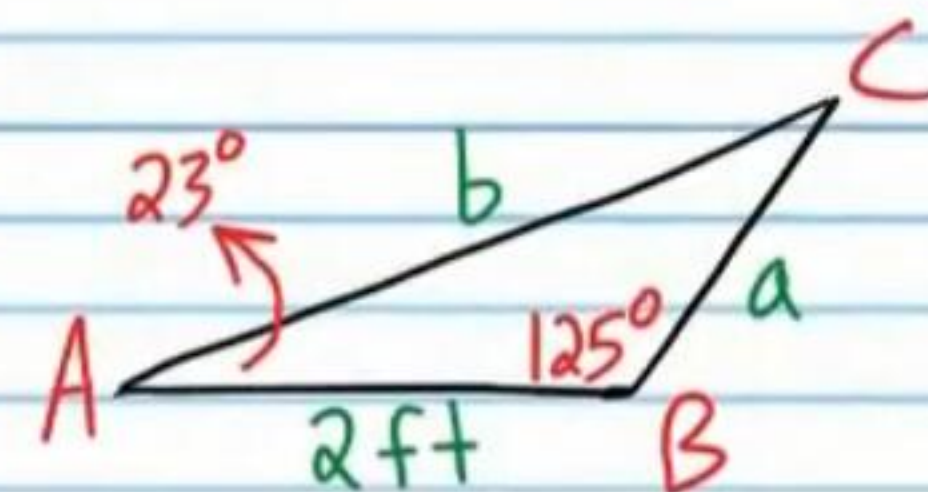


⑧ $m\angle A$.

⑨ $m\angle C$.

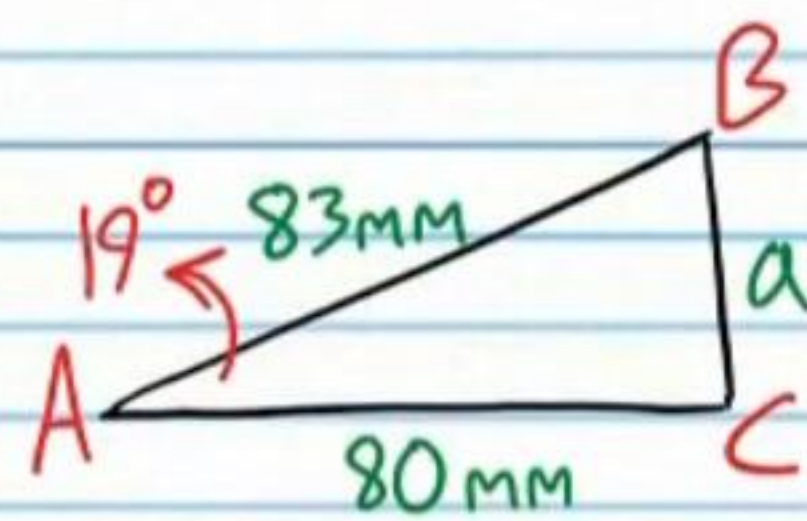
⑩ $m\angle C$

⑪ a



⑫ b

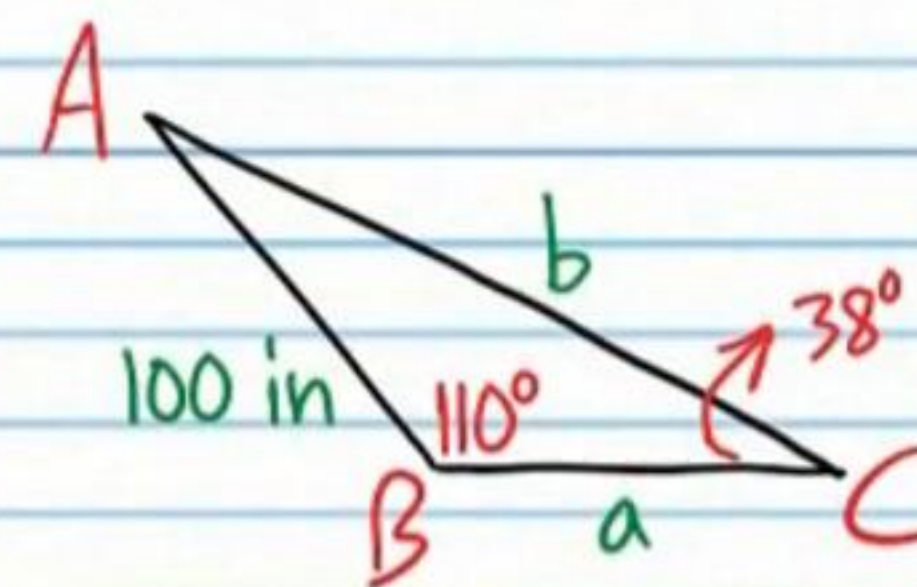
⑬ a



⑭ $m\angle C$

⑮ $m\angle B$

⑯ $m\angle A$



⑰ a

⑱ b

$$\textcircled{19} A = 50^\circ, b = 7\text{km}, a = 4\text{km}$$

②① $B = 55^\circ$, $b = 5.2 \text{ ft}$, $c = 6 \text{ ft}$

②① $C = 120^\circ$, $b = 10$ in, $c = 9$ in

$$\textcircled{22} \quad A = 100^\circ, a = 7\text{m}, c = 3\text{m}$$

For the following problems, write the equation of the midline, and find the period, amplitude, and phase shift. Then, graph the equation. *Only one cycle of the graph is necessary. Start the cycle for each graph at zero degrees.*

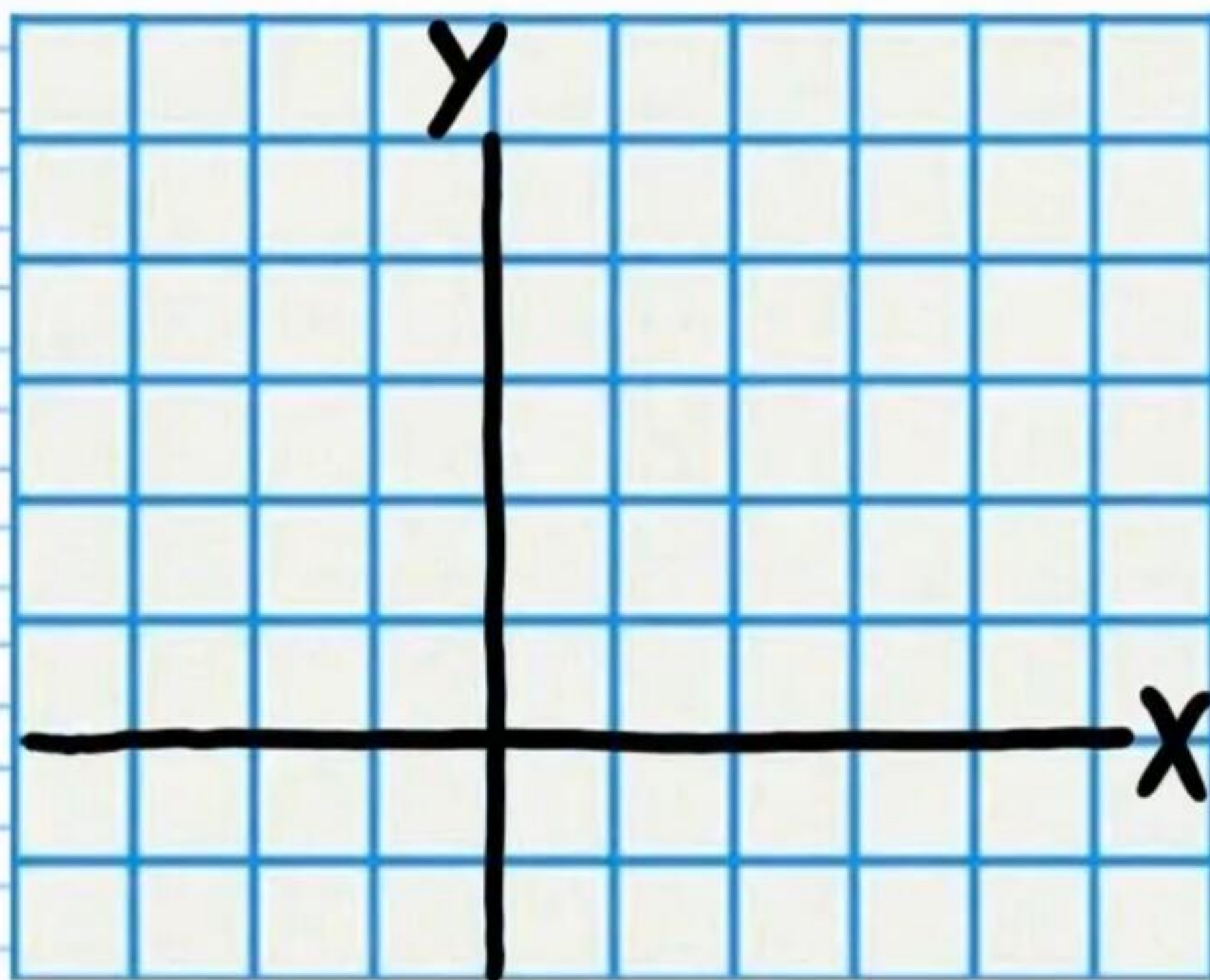
②③ $y = -2\cos(2x - 180^\circ) + 3$

Midline Equation:

Period:

Phase Shift:

Amplitude:

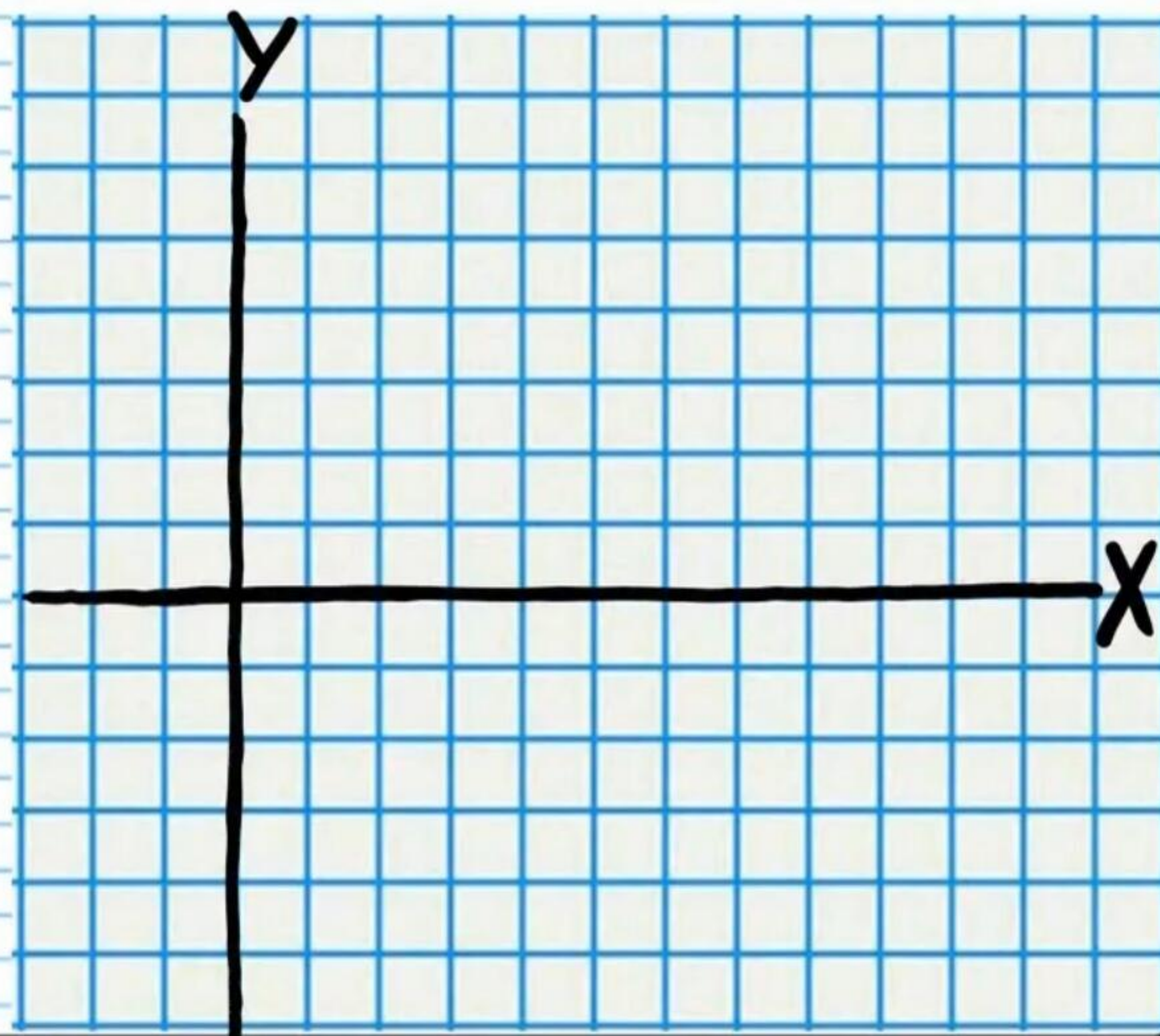


②④ $y = 2\csc\left(\frac{x}{3} + 30^\circ\right) - 2$

Midline Equation:

Period:

Phase Shift:

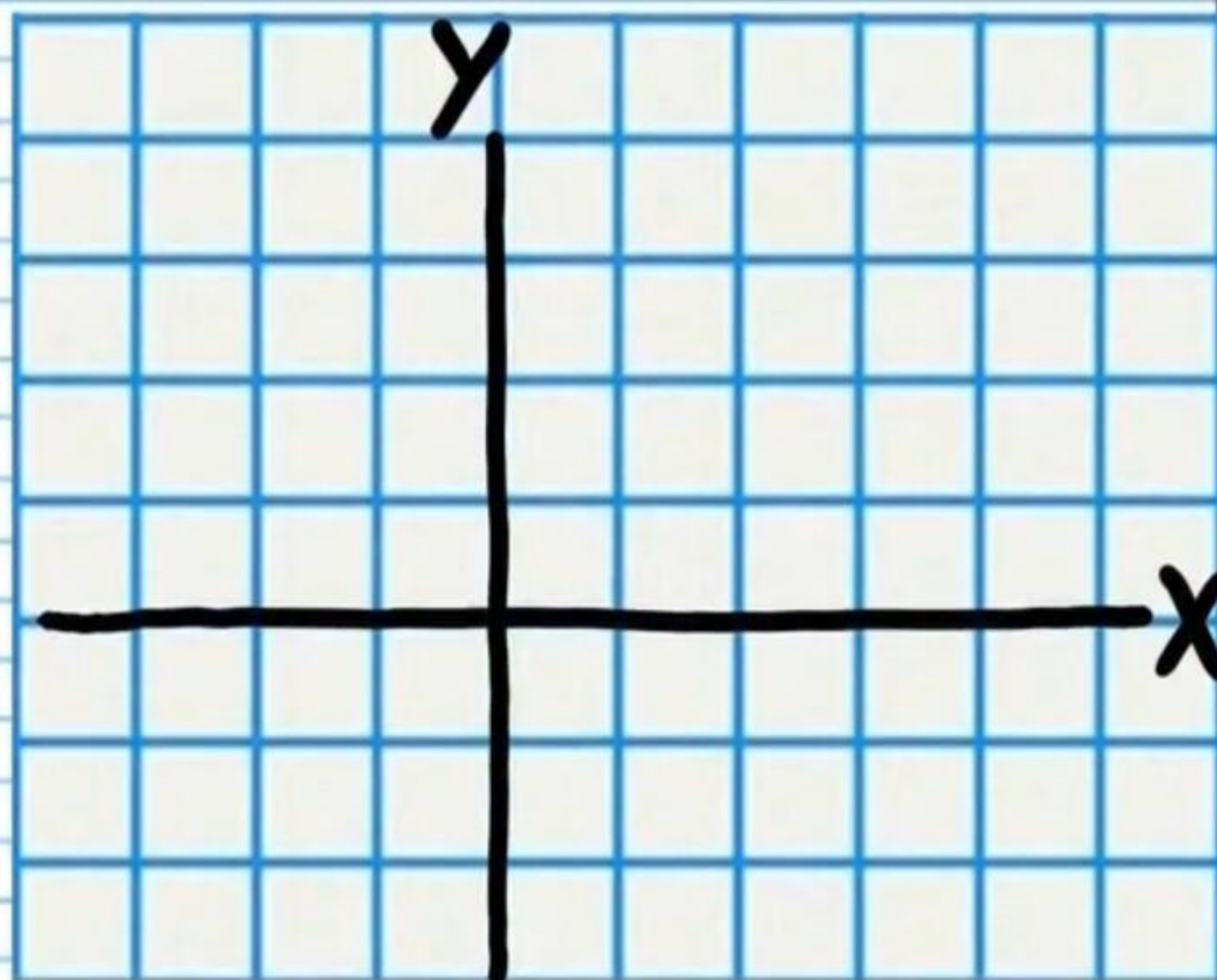


②⑤ $y = \tan(x - 180^\circ)$

Midline Equation:

Period:

Phase Shift:



Practice Test III (Version 2) Answers

① 90°

② -30°

③ -74.8°

④ $y = 35 \sin(x - 90^\circ) + 2$

⑤ $y = 3 \csc(2x - 540^\circ)$

⑥ $y = \cot(x + 90^\circ) + 1$

⑦ $B \approx 71.79^\circ$

⑧ $A \approx 58.75^\circ$

⑨ $C \approx 49.46^\circ$

⑩ $C \approx 32^\circ$

⑪ $a \approx 1.47 \text{ ft}$

⑫ $b \approx 3.09 \text{ ft}$

⑬ $a \approx 27.06 \text{ mm}$

⑭ $C \approx 86.78^\circ$

⑮ $B \approx 74.22^\circ$

⑯ $A = 32^\circ$

⑰ $a \approx 86.07 \text{ in}$

⑱ $b \approx 152.63 \text{ in}$

⑲ $\sin A \approx 1.34$

No Triangle Possible

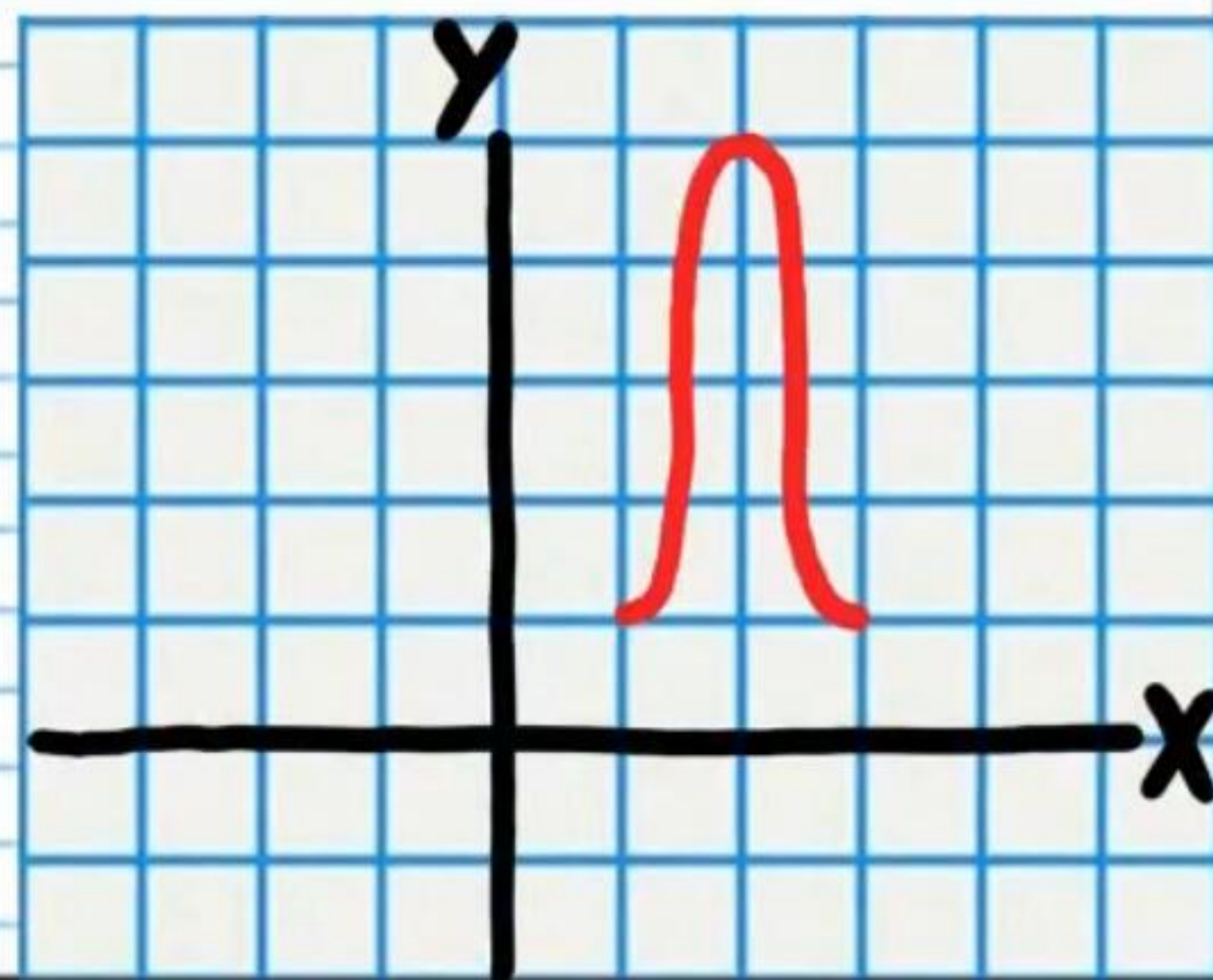
⑳ $C_1 \approx 70.94^\circ$
 $A_1 \approx 54.06^\circ$
 $a_1 \approx 5.14 \text{ ft}$
 $C_2 \approx 109.06^\circ$
 $A_2 \approx 15.94^\circ$
 $a_2 \approx 1.74 \text{ ft}$

(21) $C + B < 180^\circ$
 $120^\circ + 74.2^\circ < 180^\circ$
 $194.2^\circ < 180^\circ$

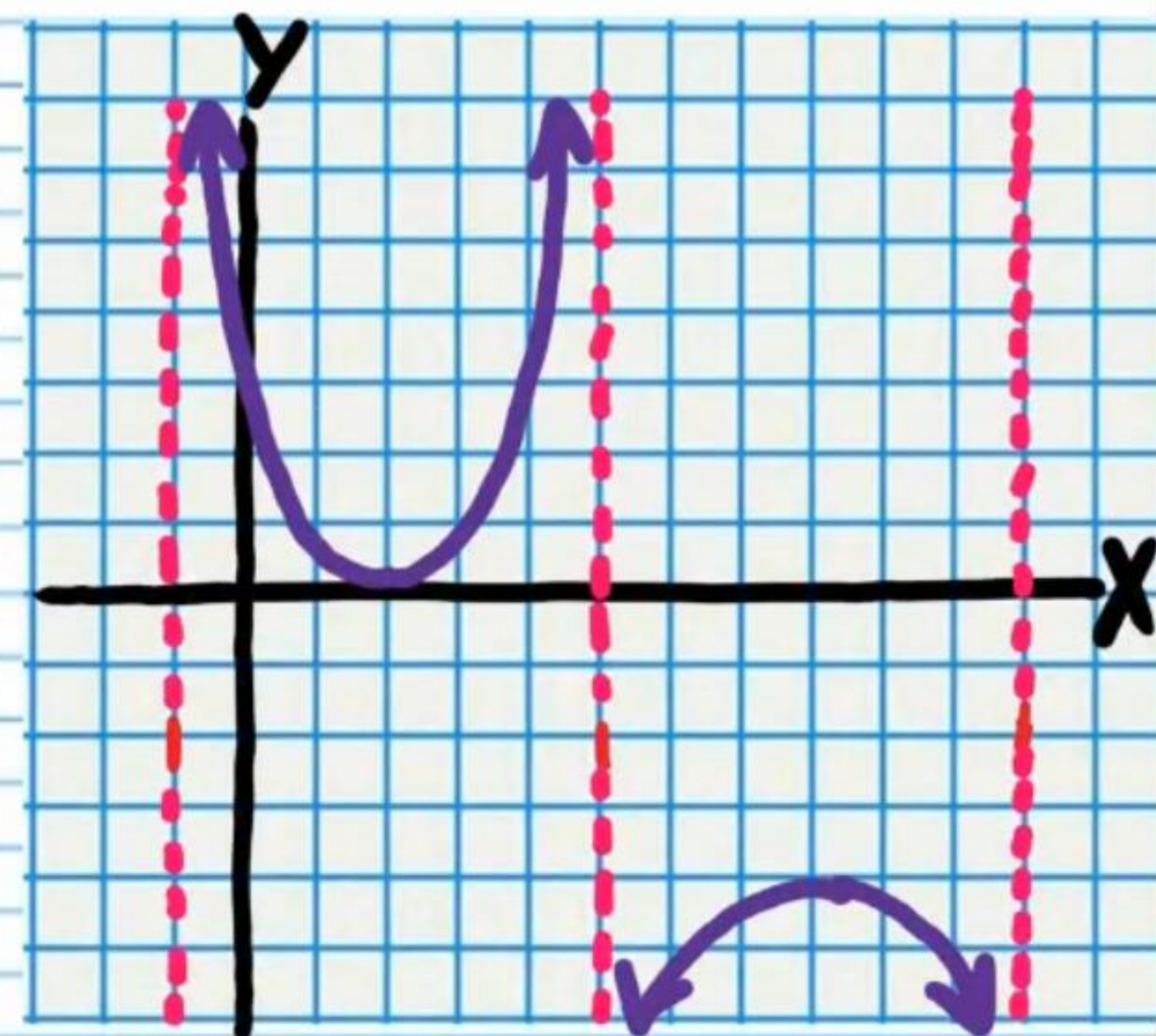
No Triangle Possible

(22) $b \approx 5.825 \text{ m}$
 $B \approx 55.035^\circ$
 $C \approx 24.965^\circ$

(23) $y = 3$
 $P = 180^\circ$
 $PS = 90^\circ$
 $A = 2$



(24) $y = -2$
 $P = 1,080^\circ$
 $PS = -90^\circ$



(25) $y = 0$
 $P = 180^\circ$
 $PS = 180^\circ$

